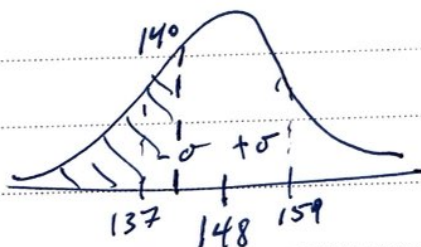


4.16 Do Now: Normal and Binomial Distribution Practice

- The heights, H cm, of sunflowers grown in a garden competition are normally distributed with a mean of 148 cm and a standard deviation of 11 cm.
 - Find $P(H < 140)$. [2]
 - Find $P(135 < H < 160)$. [2]
 - The tallest 20% of sunflowers receive a prize. Find the minimum height, in cm, required to receive a prize. [2]
 - Given that a sunflower does not receive a prize, find the probability that it is taller than 150 cm. [2]

$$\begin{aligned} a) \quad & \text{NormCDF}(0, 140, 148, 11) \\ & = 0.233579... \\ & \approx 0.234 \end{aligned}$$



$$\begin{aligned} b) \quad & \text{NormCDF}(135, 160, 148, 11) \\ & = 0.743705... \\ & \approx 0.744 \end{aligned}$$

$$\begin{aligned} c) \quad & \text{Inv NormCDF}(0.80, 148, 11) \\ & = 138.742 \quad 157.258... \\ & \approx 139 \text{ cm} \quad \approx 157 \text{ cm} \end{aligned}$$

$$\begin{aligned} d) \quad & P(150 < H < 157.258...) = \text{NormCDF}(150, 157.258..., 148, 11) \\ & = 0.227863... \end{aligned}$$

$$\begin{aligned} P(150 < H < 157.258... | H > 150) &= \frac{0.22786...}{0.8} \\ &= 0.28482... \approx 0.285 \end{aligned}$$

